Part 2: Exploratory data analysis

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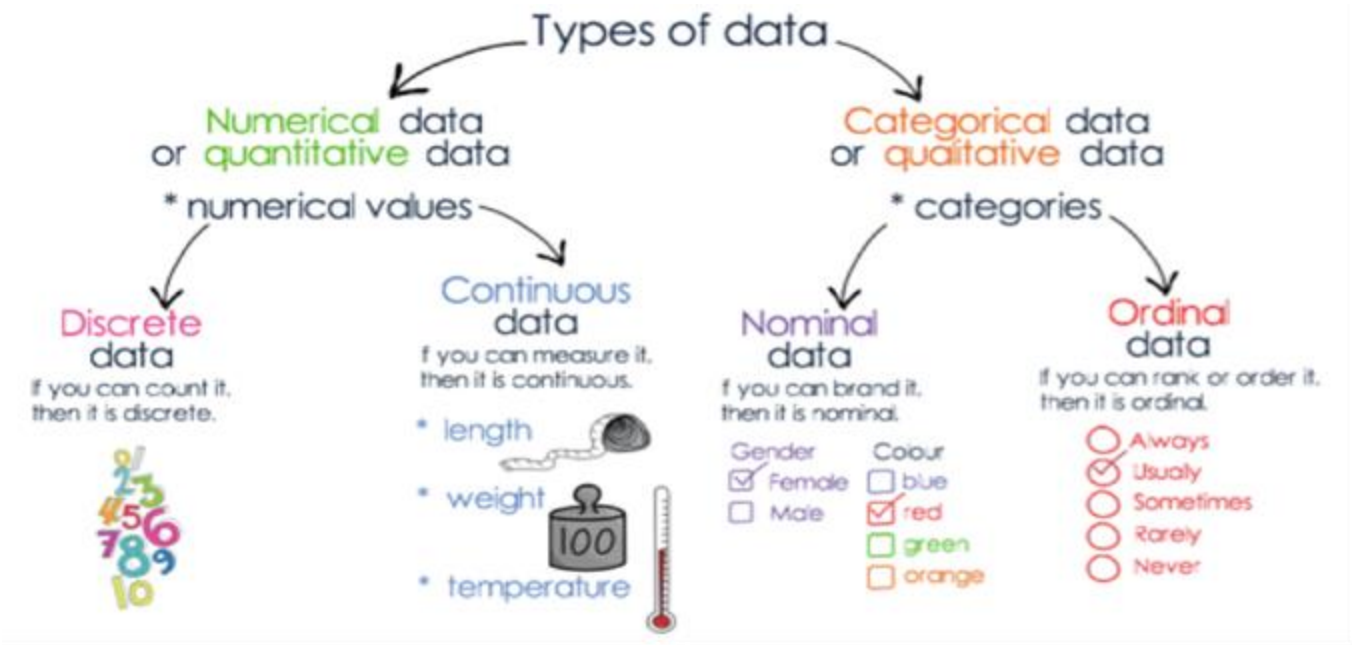
Content

1. Data types
2. Data formats
3. Plotting
4. Representative values & summary statistics
5. Results presentation and data science processes
6. Cognitive Biases

Data Types



https://legac.com.au/blogs/further-mathematics-exam-revision/further-mathematics-unit-3-data-analysis-types-of-data

Data Types: Numerical

- Arithmetic operations are possible (3 + 4 = 7)
- Enumeration is possible (3 > 2)
- Interval differences are consistent (3 – 2 = 12 – 11)
- Suitable for statistical analysis (e.g. test statistics, regression, classification)
- Visualization: Pie charts, bar charts, various plots, histograms, …

-> Numerical data is our preferred data type. If data is not numerical by nature we often make it numerical

Data Types: Numerical

Example: Weight

- 3kg + 7kg = 10kg ✓
- 3kg > 2kg ✓
- 3kg – 2kg = 10kg – 9kg ✓

Data Types: Categorical - Ordinal

- Arithmetic operations are not possible (“A lot” + “some” = ?)
- Enumeration is possible (“A lot” > “some”)
- Interval differences are not consistent (5 – 4 ≠ 3 - 2)
- Partly suitable for statistical analyses (e.g. test statistics, regression, classification)
- Visualization: Pie charts, bar charts, tables, …

-> Often, ordinal data can be converted to a numerical scale (some assumptions have to be made)

Data Types: Categorical - Ordinal

Example: Ranking

"How do you like icecream?"

- ❑ Best food ever
- ❑ Love it
- ❑ I really like it
- ❑ I like it
- ❑ It's average
- ❑ I hate it

What it possible?

- "I like it" + "I really like it" = "I love it"? ✗
- "I like it" > "It's average"? ✓
- "I love it" – "I really like it" = "It's average" – "I hate it"? ✗

Data Types: Categorical - Ordinal

Example: Ranking

"How do you like icecream?"	Translation into numerical (1)	Translation into numerical (2)
❑ Best food ever	❑ 5	❑ 10
❑ Love it	❑ 4	❑ 9
❑ I really like it	❑ 3	❑ 8
❑ I like it	❑ 2	❑ 7
❑ It's average	❑ 1	❑ 5
❑ I hate it	❑ 0	❑ 0
	Bad choice as only enumeration is achieved	Good choice as also interval differences and arithmetic operations are approx. achieved

Data Types: Categorical - Nominal

- Arithmetic operations are not possible (“Cat” + “Dog” = ?)
- Enumeration is not possible (“Dog” > “Cat” does not make sense per se)
- Interval differences are not consistent (“Cat” – “Dog” ≠ “Cat” – “Mouse”)
- Partly suitable for statistical analyses (e.g. test statistics, regression, classification)
- Visualization: Pie charts, bar charts, tables, …

-> Nominal data is often converted to Numerical data by One-hot-encoding

Data Types: Categorical - Nominal

Example: Multiple choice

"What's your favorite animal?"

- ❑ Dog
- ❑ Cat
- ❑ Monkey
- ❑ Elephant

What is possible?

- "Elephant" + "Monkey" = "Cat"? ✗
- "Cat" > "Monkey"? ✗
- "Dog" – "Cat" = "Monkey" – "Elephant"? ✗

One-hot-encoding

Technique to transform categorical into numerical features

Feature f = {Cat; Dog; Mouse}

Feature f_cat = {0; 1}
Feature f_dog = {0; 1}
Feature f_mouse = {0; 1}

Example:

Example	f
1	“Cat”
2	“Dog”
3	“Mouse”

Example	f_cat	f_dog	f_mouse
1	1	0	0
2	0	1	0
3	0	0	1

Structured vs unstructured data

- **structured data** are well defined and highly organized. It is easy to find what you want with a classic SQL request in RDBMS such as MS SQL Server, MySQL, and Microsoft Access.

- **unstructured data** are not structured and can be anything that’s not in any specific format, e.g. audio format, video format, posts from twitter, emails

- Both data types typically need preprocessing/feature extraction

Data format: CSV

- A **CSV** (**c**omma **s**eparated **v**alues) file stores tabular data.
- Each line of the file represent a data record.
- Each record consists of one or more fields.

```
movieId,title,genres
1,Toy Story (1995),Adventure|Animation|Children|Comedy|Fantasy
2,Jumanji (1995),Adventure|Children|Fantasy
3,Grumpier Old Men (1995),Comedy|Romance
4,Waiting to Exhale (1995),Comedy|Drama|Romance
```

- The CSV file format is not fully standardized. When the data contain commas, another delimiter may be used, for e.g. a semicolon ‘;’.

Data format: JSON

- JSON (JavaScript Object Notation) format consists of key/value pairs.
- The key and the value is separated with a colon “:”.
- A key-value pair is separated from another pair by a comma.
- Each entry is enclosed by curly brackets.
- JSON is standardized.

```
1  [
2    {
3      "movieId": 1,
4      "title": "Toy Story (1995)",
5      "genres": "Adventure|Animation|Children|Comedy|Fantasy"
6    },
7    {
8      "movieId": 2,
9      "title": "Jumanji (1995)",
10     "genres": "Adventure|Children|Fantasy"
11   },
12   {
13     "movieId": 3,
14     "title": "Grumpier Old Men (1995)",
15     "genres": "Comedy|Romance"
16   },
17   {
18     "movieId": 4,
19     "title": "Waiting to Exhale (1995)",
20     "genres": "Comedy|Drama|Romance"
21   }
22 ]
```

Data Frames

- A ***data frame*** is a table or a 2D array-like structure that contains data that might be of different types
  - each column contains values of one variable
  - each row contains one set of values from each column
- Data wrangling and visualization in Python can be done with **pandas** package.

pandas.pydata.org

*"**pandas** is a powerful, flexible and easy to use open source data analysis and manipulation tool, built on top of the Python programming language."*

- A data frame can be created from a dictionary or by importing a file, for e.g. a .CSV file.
  - Other data formats: https://pandas.pydata.org/pandas-docs/stable/reference/io.html

Exploring Data Frames

- Subsetting columns:
  - by the name or multiple names in a list

```
ratings_df[["movieId", "rating"]].head()
```

	movieId	rating
0	1	4.0
1	3	4.0
2	6	4.0
3	47	5.0
4	50	5.0

- Subsetting rows using a Boolean series

```
# This will result in a Boolean series
ratings_df["userId"] == 1
```

```
0          True
1          True
2          True
3          True
4          True
          ...
100831    False
100832    False
100833    False
100834    False
100835    False
Name: userId, Length: 100836, dtype: bool
```

```
#Filtering for rows where userId=1
ratings_df[ratings_df["userId"]==1]
```

	userId	movieId	rating	time
0	1	1	4.0	2000-07-30 18:45:03
1	1	3	4.0	2000-07-30 18:20:47
2	1	6	4.0	2000-07-30 18:37:04
3	1	47	5.0	2000-07-30 19:03:35
4	1	50	5.0	2000-07-30 18:48:51
...	...	...	...	...
227	1	3744	4.0	2000-07-30 18:11:34
228	1	3793	5.0	2000-07-30 18:30:55
229	1	3809	4.0	2000-07-30 18:20:20
230	1	4006	4.0	2000-07-30 18:48:23
231	1	5060	5.0	2000-07-30 19:06:42

Exploring Data Frames

- Subsetting rows using `.isin()`

```
# also results in Boolean series
is_movie123 = ratings_df['movieId'].isin([1, 2, 3])

#subset ratings_df with is_movie123
ratings_df[is_movie123]
```

	userId	movieId	rating	time
0	1	1	4.0	2000-07-30 18:45:03
1	1	3	4.0	2000-07-30 18:20:47
516	5	1	4.0	1996-11-08 06:36:02
560	6	2	4.0	1996-10-17 11:58:42
561	6	3	5.0	1996-10-17 12:11:36
...	...	...	...	...
98666	608	1	2.5	2005-05-29 23:11:07
98667	608	2	2.0	2005-05-30 22:06:26
98668	608	3	2.0	2005-05-31 01:53:33
99497	609	1	3.0	1996-11-05 19:10:25
99534	610	1	5.0	2016-11-19 08:08:20

- Subsetting rows using **indices** by `loc` (label) and `iloc` (integer position).

```
tags_df.iloc[[2,0]]
```

	userId	movieId	time
tag			
will ferrell	2	60756	2015-10-24 19:29:52
funny	2	60756	2015-10-24 19:29:54

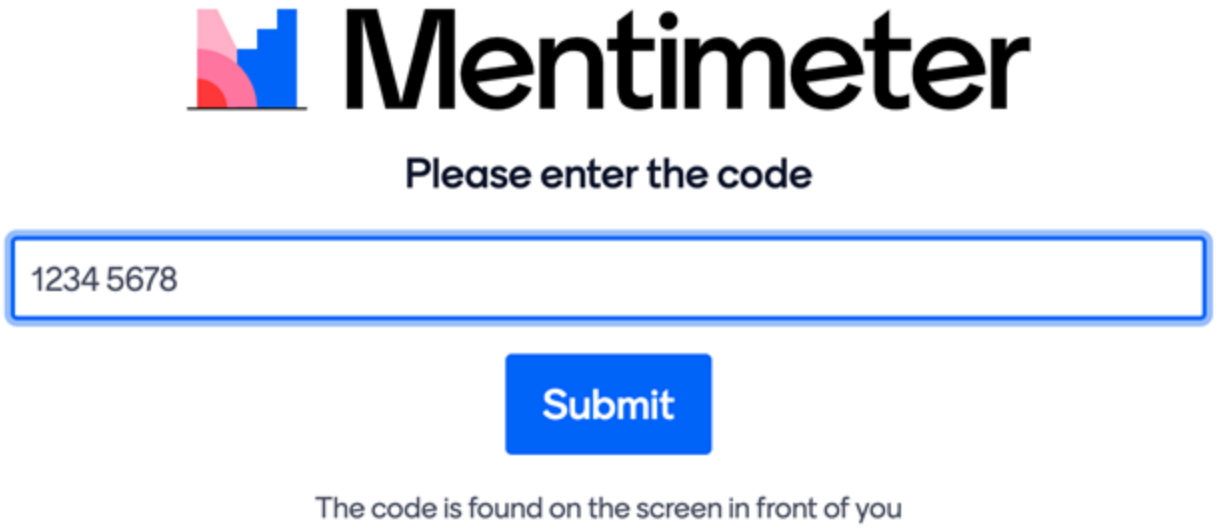
```
tags_df = tags_df.set_index('tag')
tags_df.head()
```

	userId	movieId	time
tag			
funny	2	60756	2015-10-24 19:29:54
Highly quotable	2	60756	2015-10-24 19:29:56
will ferrell	2	60756	2015-10-24 19:29:52
Boxing story	2	89774	2015-10-24 19:33:27
MMA	2	89774	2015-10-24 19:33:20

```
tags_df.loc[['funny', 'will ferrell']].tail()
```

	userId	movieId	time
tag			
funny	599	1732	2017-06-26 05:51:31
will ferrell	2	60756	2015-10-24 19:29:52
will ferrell	62	60756	2018-06-13 23:59:39
will ferrell	62	107348	2018-06-14 00:10:02
will ferrell	424	60756	2016-03-13 05:15:29

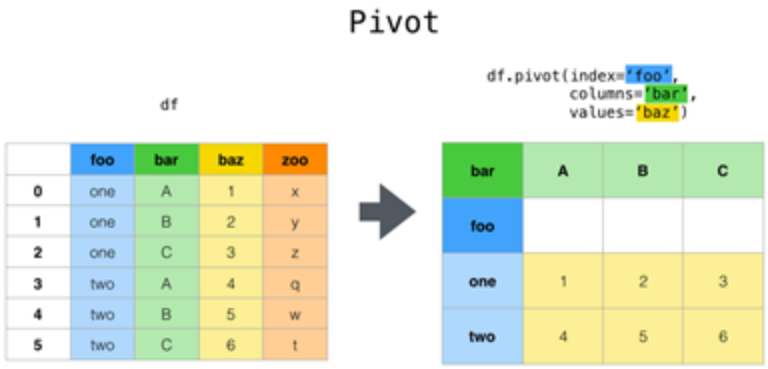
Mentimeter


Mentimeter
Please enter the code
1234 5678
Submit
The code is found on the screen in front of you

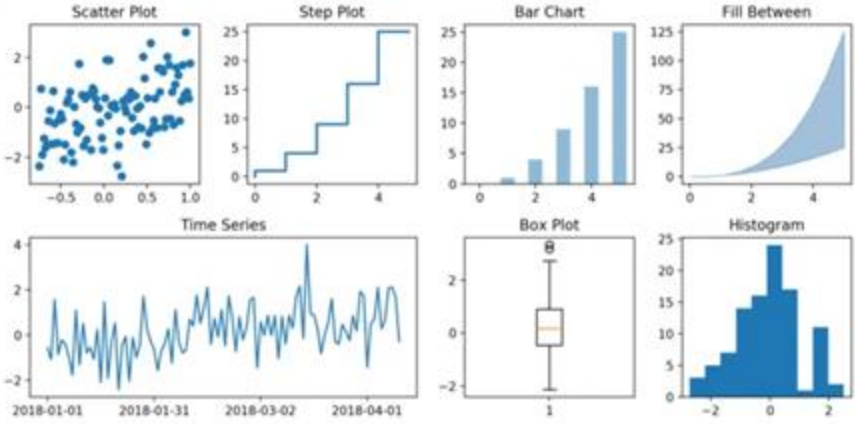
Plotting

Why data exploration

- We want to understand data properties and patterns in the data before “diving in”.
- Understanding the data allow us to plan the modeling strategies.
- To effectively communicate said strategies and the results in our report.
- Graphs/plots reveal structure that summaries don’t.



https://pandas.pydata.org/pandas-docs/stable/user_guide/reshaping.html



https://www.machinelearningplus.com/plots/matplotlib-tutorial-complete-guide-python-plot-examples/

Principles of EDA

1. Avoid ridiculograms

   "**ridiculogram**: meaningless albeit visually impressive image of a network"

1. Show the data
   1. Use the appropriate plot/table
   2. Your plot/table should be clear
2. Compare things directly

   ❌ use two graphs ✅ plot on the same graph ✅ plot (log) ratio

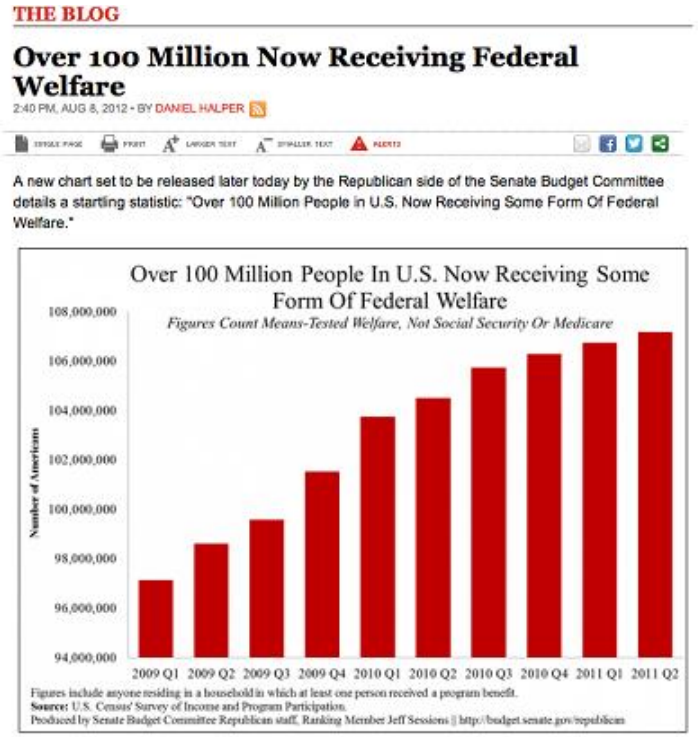
1. Be careful with scale:
   1. use common scales
   2. consider starting at zero

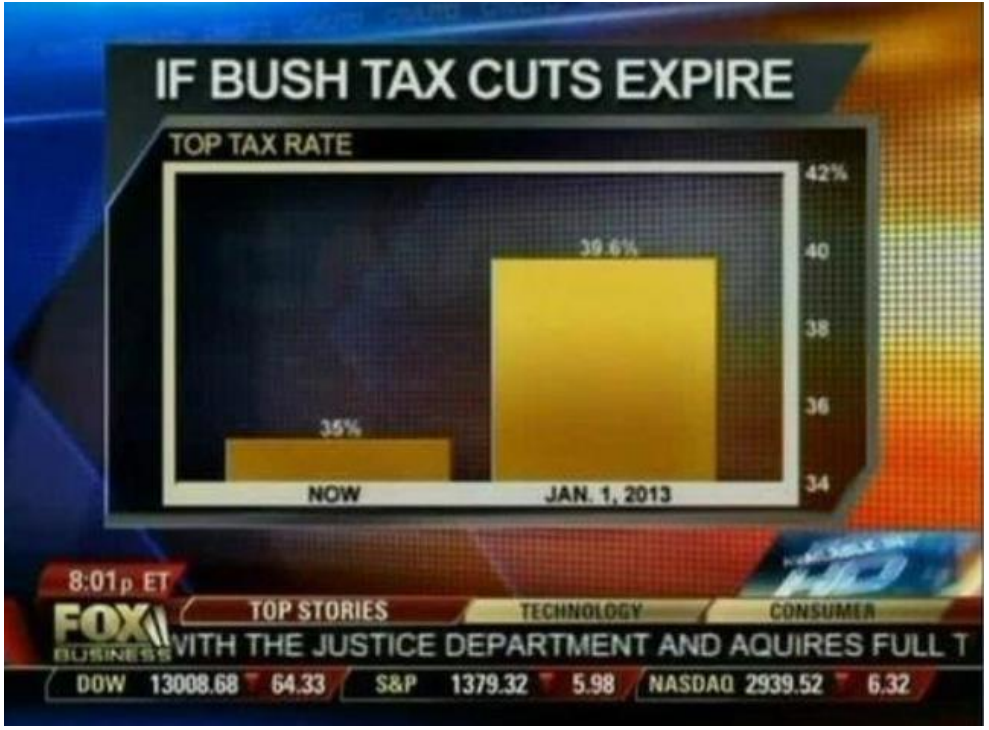
morethanmyheight.com, jtleek.com

Most common plotting mistakes

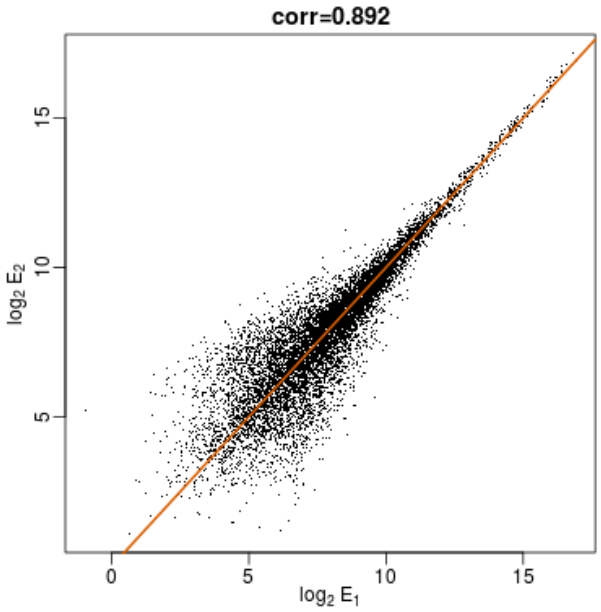
- No proper axes description
- Comparing without using the same scale for the x- or y-axis
- No display of uncertainty/error bars if appropriate
- Inconsistent use of colors
- Too much information in one plot: A plot is a storyteller. Let it tell a good story that is immediately clear to everyone

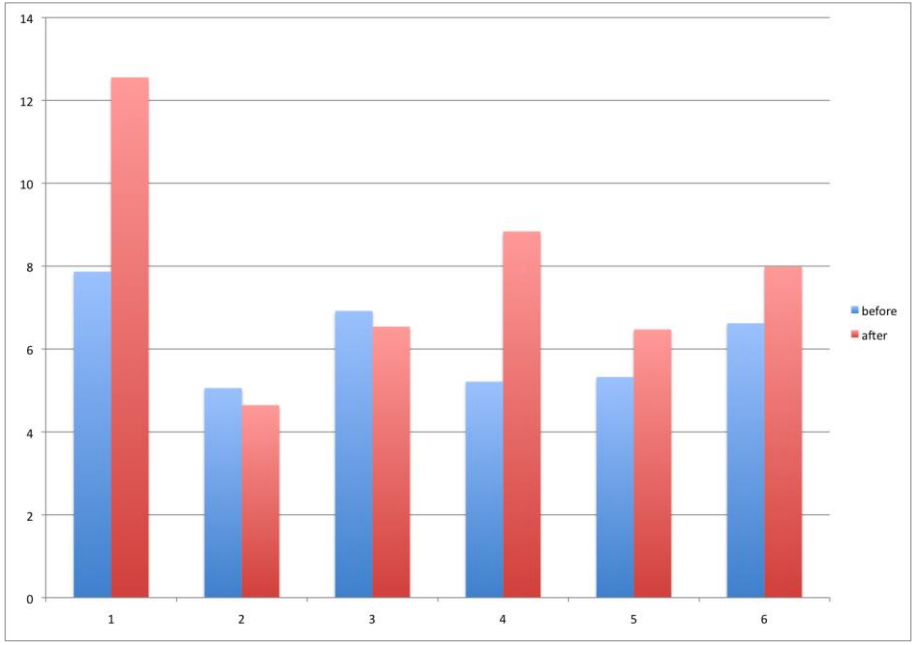
Examples of bad plots



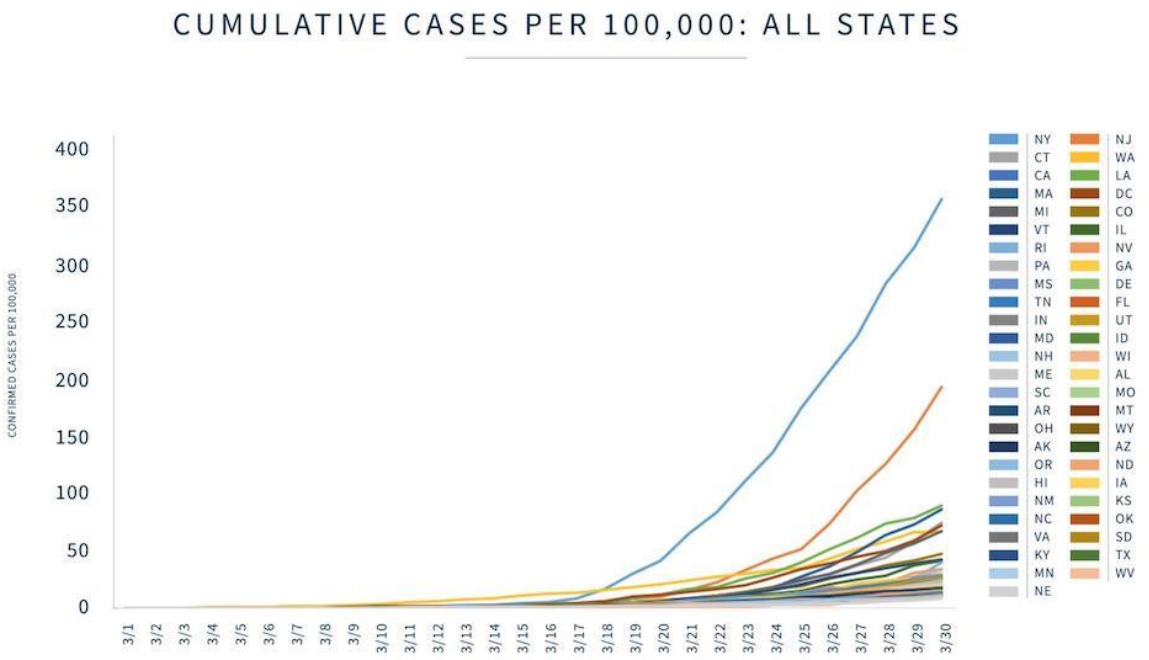


Examples of bad plots



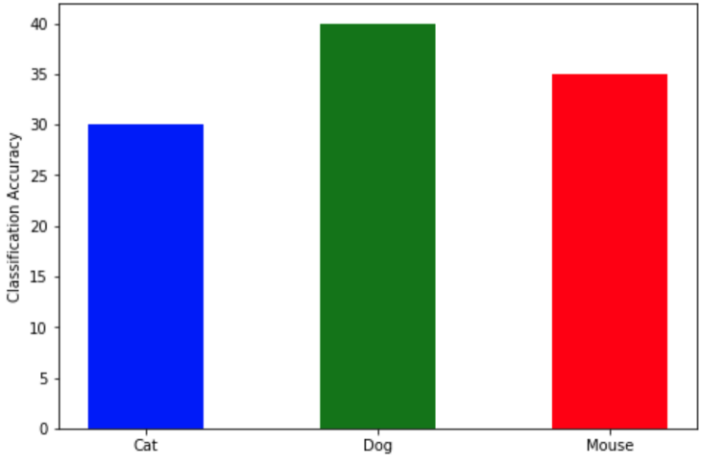


Examples of bad plots

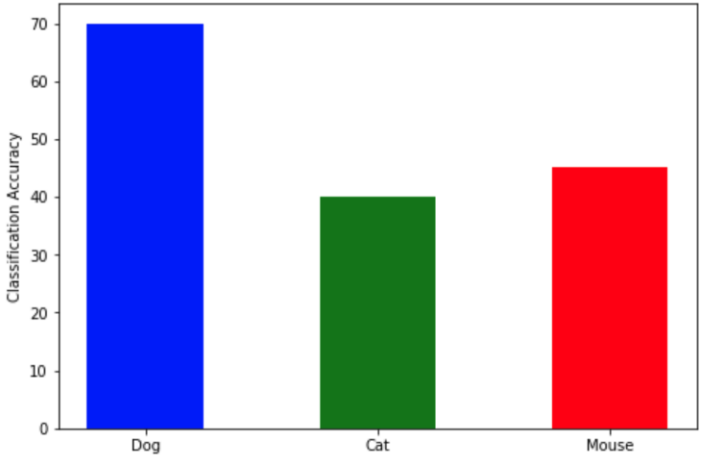


Examples of bad plots

Before



After



- In practically every plotting tool the axis range gets determined automatically
  -> If possible show in the same plot, otherwise keep all axes in the same range
- Color coding is often according to order, not content. You have to manage your own color consistency

Matplotlib and Seaborn

- We can produce basic plots with the Matplotlib package.
- Seaborn is a visualization library based on Matplotlib that can produce elegant plots with fewer syntaxes.

In [1]:

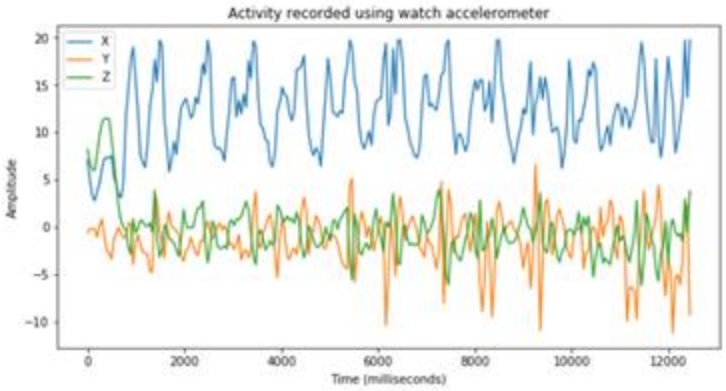
```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('accel_watch.csv').set_index('timestamp')
df.head(3)
```

Out[1]:

	X	Y	Z
timestamp			
0.0	7.091625	-0.591667	8.195502
49.5	4.972757	-0.158317	6.696732
99.0	3.253720	-0.191835	6.107758

In [2]:

```python
plt.figure(figsize=(10,5))
plt.plot(df)
plt.title('Activity recorded using watch accelerometer')
plt.xlabel('Time (milliseconds)')
plt.ylabel('Amplitude')
plt.legend(('X', 'Y', 'Z'))
plt.show()
```



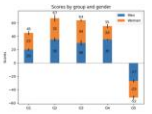
Matplotlib reference

Gallery

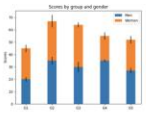
This gallery contains examples of the many things you can do with Matplotlib. Click on any image to see the full image and source code.

For longer tutorials, see our tutorials page. You can also find external resources and a FAQ in our user guide.

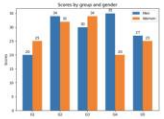
Lines, bars and markers



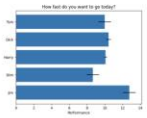
Bar Label Demo



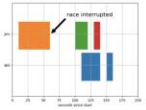
Stacked bar chart



Grouped bar chart with labels

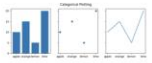


Horizontal bar chart

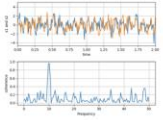


Broken Barh

CapStyle



Plotting categorical variables



Plotting the coherence of two signals

https://matplotlib.org/stable/gallery/index.html

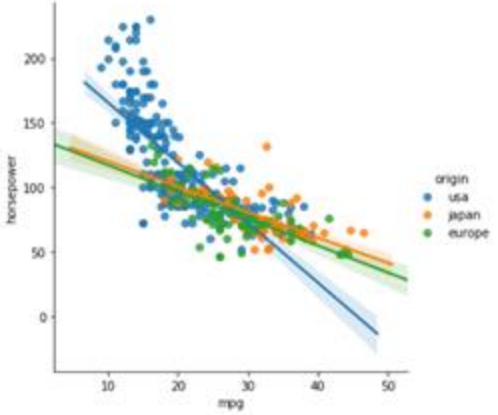
Seaborn example: Auto MPG dataset

In [4]:
```
mpg_df = sns.load_dataset('mpg')
mpg_df.head(2)
```

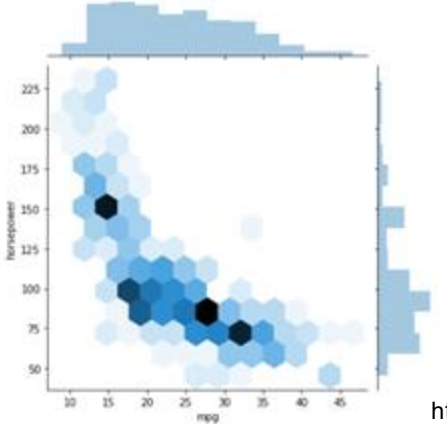
Out[4]:

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	name
0	18.0	8	307.0	130.0	3504	12.0	70	usa	chevrolet chevelle malibu
1	15.0	8	350.0	165.0	3693	11.5	70	usa	buick skylark 320

In [5]:
```
# using hue by origin
sns.lmplot(x='mpg', y='horsepower', hue='origin', data=mpg_df)
plt.show()
```

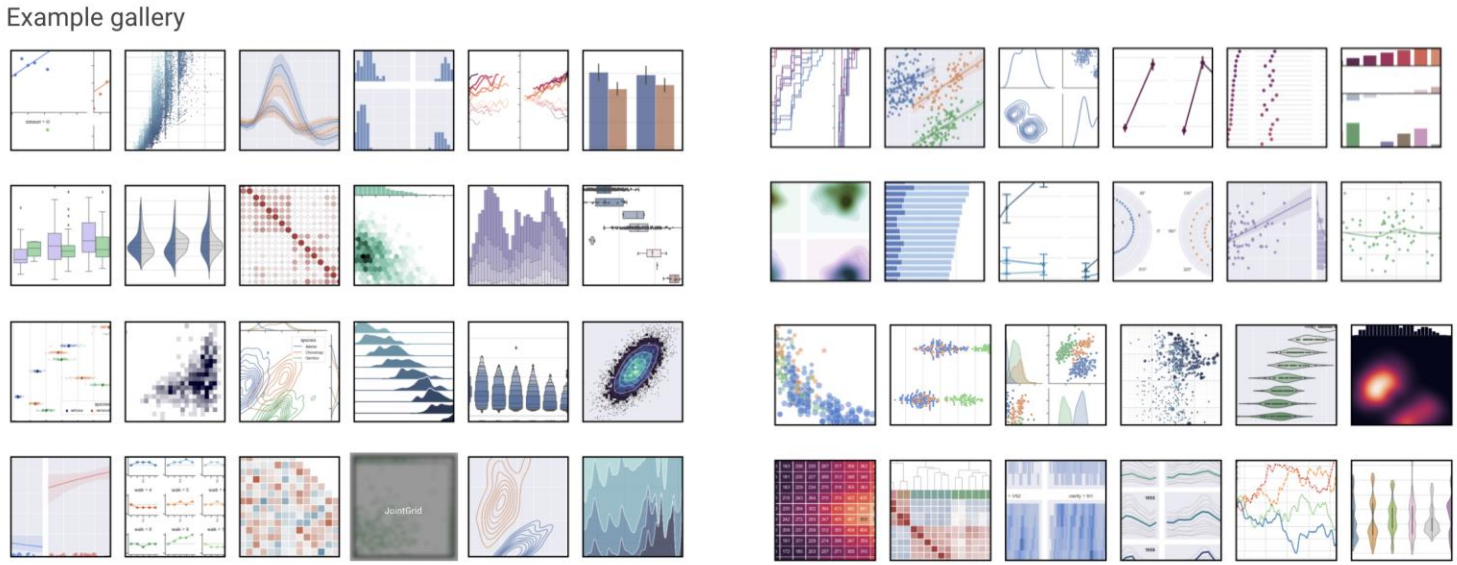


In [6]:
```
# Jointplot kind="scatter" | "reg" | "resid" | "kde" | "hex"
sns.jointplot(x='mpg', y='horsepower', kind='hex', data=mpg_df)
plt.show()
```



https://seaborn.pydata.org/examples/index.html

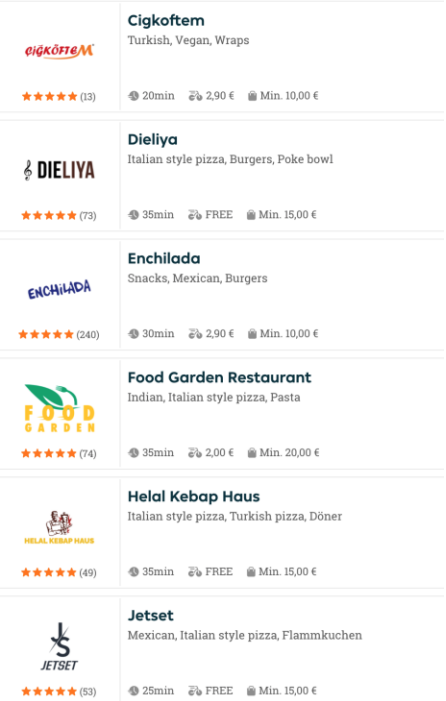
Seaborn gallery



https://seaborn.pydata.org/examples/index.html

Representative values & summary statistics

Representative values: Mean and median



From: lieferando.de

Statistical: The sample mean is the maximum likelihood estimator of the mode of a Gaussian distribution

Storytelling: The sample mean is a representative value that can summarize complexity and help to make decisions

Problems and things to consider when using the sample mean

- Its certainty depends on the number of underlying samples
  -> Use together with standard deviation or percentiles
- It is not robust
  -> Consider using the median
- It is often oversimplifying complex relationships
  -> Start with mean or median, then go more complex if needed (Stratification, correlations, distributions)

Representative values: Standard deviation and standard error

Two important metrics:

1. **Standard deviation** as a measure of variability of your data

   Two restaurants might have the same average rating of 3/5, one because it is really an average restaurant (everyone votes 3/5), the other because it is a “hate it or love it” restaurant half of the people vote 1/5, the other have 5/5

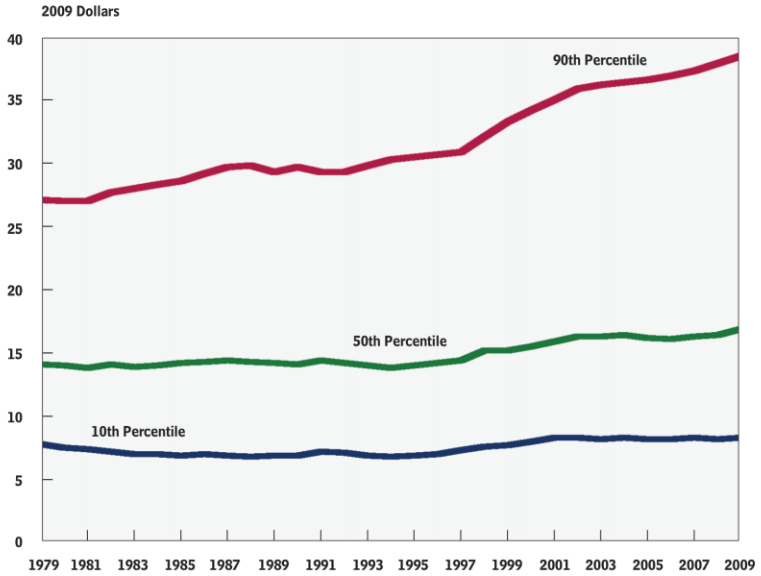
   -> They will differ in terms of standard deviation

2. Standard error of the mean. Our sample mean is an estimator whose quality depends on the underlying data, most importantly the number of samples

Two restaurants might have the same average rating of 5/5 and even the same standard deviation of almost 0. One might be based on 5 ratings, the other based on 5000 ratings

Standard error of the mean: $\frac{\sigma}{\sqrt{n}}$

Percentiles



Hourly Wages at Selected Percentiles for Workers Ages 16 to 64

A mean or median value is highly aggregated and can be misleading

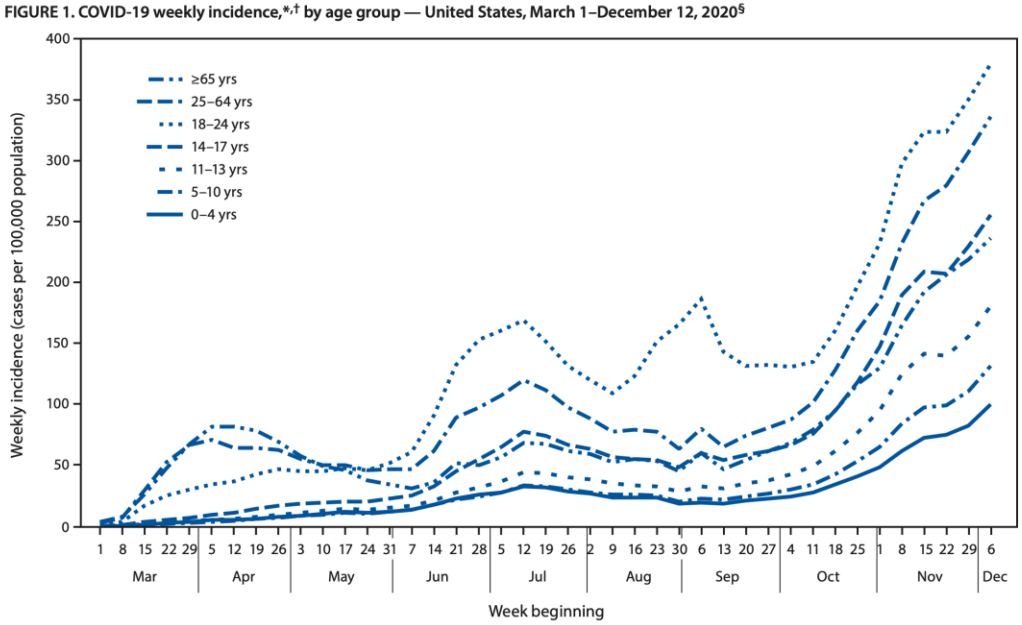
A first good step is to look at percentiles

The chart on the left side shows that

- In average the hourly wage increased
- But: For the average worker it stayed constant
- The average increase is merely representing the increase that high-earning people experienced

From: https://www.cbo.gov/sites/default/files/cbofiles/ftpdocs/120xx/doc12051/02-16-wagedispersion.pdf

Stratification



Stratification is the process of classifying your data into different groups to do separate analyses. Typical stratifications are

- Age
- Gender
- Ethnicity
- Income
- Education
- …

From: https://www.cdc.gov/mmwr/volumes/70/wr/pdfs/mm7003e1-H.pdf

From mean to more

Example: Development of wages

- Start by observing mean values, e.g. over time
  -> Conclusion: The average wage increased significantly over time

- Fix your data for any distortions (here: inflation)
  -> Conclusion: It still increases over time, but only a bit

- Look at the percentiles
  -> Conclusion: Actually it increased a lot for high-wages and stayed constant for the majority

- Stratification
  -> Conclusion: Actually it increased significantly for white males. For other groups it even decreased

- Too much stratification
  -> Conclusion: The average wage increased by 200% for 38 year old females from Darmstadt who are dog and cat owners, have a degree in electrical engineering and have more than 1000 Instagram followers
  Sample size n = 3 -> Large standard error on the average wage for this group

Percentage vs percentage points

One of the most common confusions when communicating results:

Example: The VAT in Germany was reduced from 19% to 16% in 2020

Option 1: “VAT got reduced by 3 percentage points”

Option 2: “VAT got reduced by 15.8%”

Option 3: “Prices got 2.5% lower” (from 1,19€ to 1,16€)

Data Science example: My classifier went from 30% to 60% accuracy

Option 1: “Its accuracy increased by 30 percentage points”

Option 2: “It is 100% more accurate”

Option 3: “The error rate went down by 30 percentage points”

Option 4: “The error rate went down by 42.9%”

Percentage vs percentage points

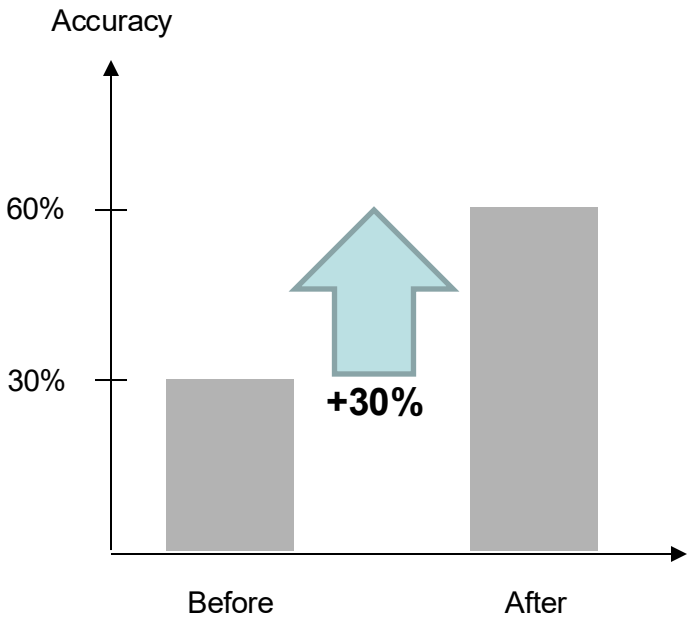
Data Science example: My classifier went from 30% to 60% accuracy

Option 1: “Its accuracy increased by 30 percentage points”

Option 2: “It is 100% more accurate”

Option 3: “The error rate went down by 30 percentage points”

Option 4: “The error rate went down by 57%”



Percentage vs percentage points

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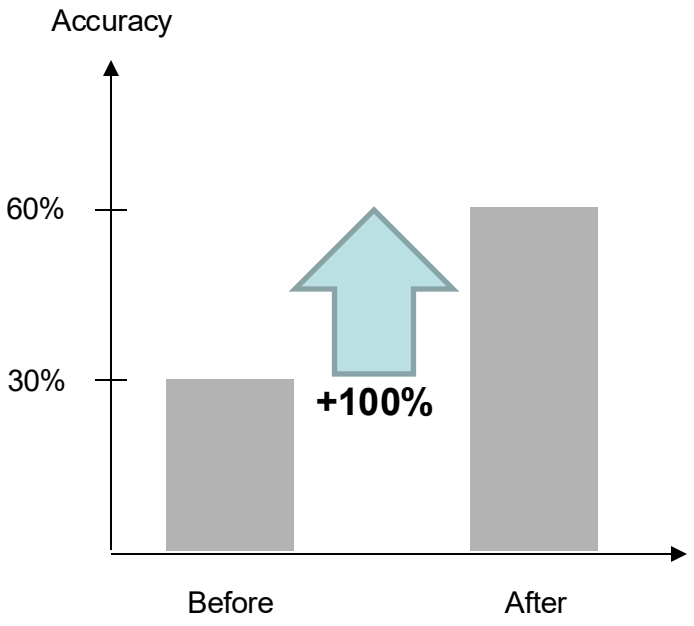


Summary statistics

- Summarizing numerical data
  - mean, median, mode, min, max, variance, standard deviation, percentiles/quantiles, summation.
- Summarizing dates
  - earliest, latest, duration
- Counting categorical variables

In [1]:
```
import pandas as pd
ratings_df = pd.read_csv('ml-latest-small/ratings.csv')
ratings_df["time"] = pd.to_datetime(ratings_df.timestamp, unit='s')
ratings_df = ratings_df.drop(columns="timestamp")
ratings_df.head(2)
```

Out[1]:

	userId	movieId	rating	time
0	1	1	4.0	2000-07-30 18:45:03
1	1	3	4.0	2000-07-30 18:20:47

In [2]:
```
# Show the earliest & the last ratings
print(ratings_df.time.min())
print(ratings_df.time.max())
```

```
1996-03-29 18:36:55
2018-09-24 14:27:30
```

In [3]:
```
# summary statistics for all ratings
ratings_df.rating.agg(['min', 'max', 'mean', 'median', 'count'])
```

Out[3]:
```
min            0.500000
max            5.000000
mean           3.501557
median         3.500000
count     100836.000000
Name: rating, dtype: float64
```

Grouped summary statistics

- With Pandas, it is easy to group summary statistics based on a certain variable.

In [4]:
```
# summary statistics of each movie
ratings_df.groupby("movieId").rating.agg(['min', 'max', 'mean', 'median', 'count'])
```

Out[4]:

movieId	min	max	mean	median	count
1	0.5	5.0	3.920930	4.0	215
2	0.5	5.0	3.431818	3.5	110
3	0.5	5.0	3.259615	3.0	52
4	1.0	3.0	2.357143	3.0	7
5	0.5	5.0	3.071429	3.0	49
...	...	...	...	...	...
193581	4.0	4.0	4.000000	4.0	1
193583	3.5	3.5	3.500000	3.5	1
193585	3.5	3.5	3.500000	3.5	1
193587	3.5	3.5	3.500000	3.5	1
193609	4.0	4.0	4.000000	4.0	1

9724 rows × 5 columns

Merging and Pivoting

```python
# merge two DataFrames on two columns, "userId" and "movieId":
tags_ratings_df = tags_df.merge(ratings_df, on=["userId", "movieId"])
tags_ratings_df["year"] = tags_ratings_df.time.dt.year
tags_ratings_df
```

	index	tag	userId	movieId	rating	time	year
0	0	funny	2	60756	5.0	2015-10-24 19:29:40	2015
1	1	Highly quotable	2	60756	5.0	2015-10-24 19:29:40	2015
2	2	will ferrell	2	60756	5.0	2015-	
3	3	Boxing story	2	89774	5.0	2015-	
4	4	MMA	2	89774	5.0	2015-	
...	...	...	...	...	...		
3471	3677	World War II	606	6107	4.0	2007-	
3472	3678	for katie	606	7382	4.5	2007-	
3473	3680	gun fu	610	3265	5.0	2016-	
3474	3681	heroic bloodshed	610	3265	5.0	2016-	
3475	3682	Heroic Bloodshed	610	168248	5.0	2017-	

3476 rows × 7 columns

superhero genre didn’t seem to be doing well in years 2003 and 2015!

```python
# Pivot table by year and aggregating the ratings with their average
tags_pivot = tags_ratings_df.pivot_table(index="tag", columns="year",
                                         values="rating", aggfunc="mean")

# sort by least occurrence of NAs, and replace NAs by blanks
tags_pivot.iloc[tags_pivot.isnull().sum(axis=1).argsort()].fillna("").head()
```

year / tag	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
music	5	3.5		2.5	4	4		4	5	4.33333						4.25	3	5	
psychology		4			3.5	4			5	4				5		5	4.41667	5	4.75
superhero		4		2.6	4.625	4							4.5	5		1.5			3.57143
time travel			3	3.5	4.33333	4.16667			4.5	4.5						4	3.5		
twist ending		4		4					4.25				4.5				4.66667	5	5

Results presentation and data science processes

Results presentation for data scientists

Assume a problem can best be solved with algorithm A, providing 90% accuracy

Data Scientist 1

Manages to find algorithm A

Has trouble relating the results to the business and communicate with non-technical people

No further investment/algorithm doesn’t go to production

0% impact

Data Scientist 2

Finds algorithm B with 70% accuracy

Relates the results to the business and communicates well with non-technical people

Further investment/algorithm goes to production

70% impact

Typical data science process

Business problem → Proof of value → MVP → Product Definition

Business problem

Who is our user? Who is the stakeholder?

What are their pain points?

What is the potential impact?

Proof of value

Is it possible to solve the problem?

Data quality, data availability, technical complexity

What performance can be expected?

MVP

Minimal viable product

How would our customers use the solution?

Stand-alone? Embedded? User interface? Process?

Product Definition

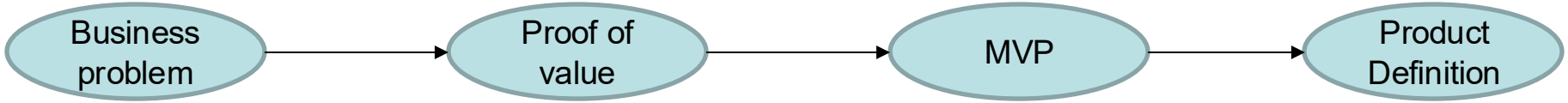
Clear product vision

Requirements, user stories, personas…

Roadmap & Budget

Clear business case

Typical data science process



What data do I need?

What approvals do I need?

Which algorithms do I use?

Code!

-> example: 2 weeks

Understand my users / feel their pain points

Identify core features

Build / Hack a version that is usable

-> example: 2 months

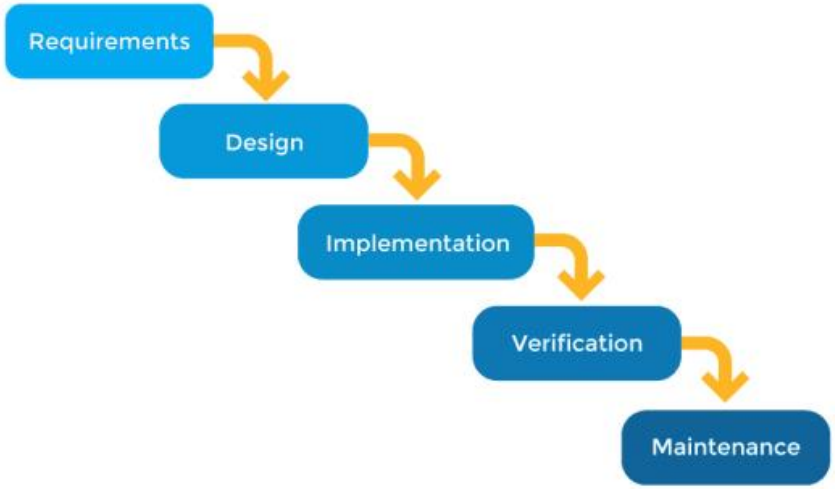
User survey / shadowing

What skills do I need to build the data science part?

What are the processes? Technical and non-technical

-> example: 1-2 months

Typical data science process



From: project-management.com

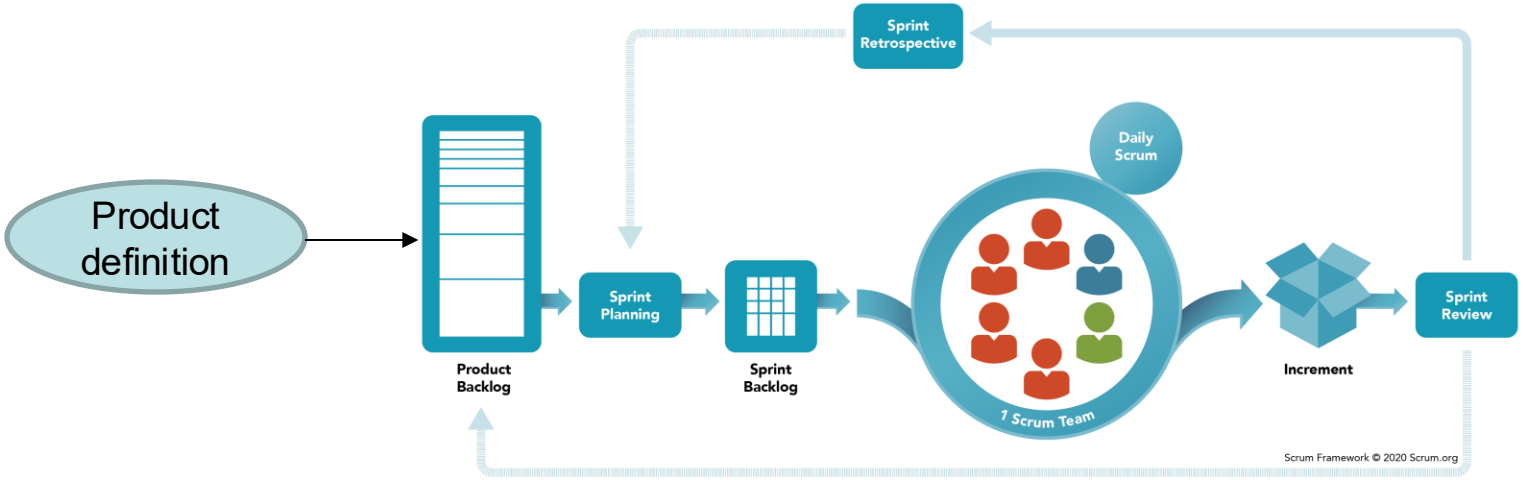
In former times (and sometimes still today) a so-called waterfall project management was used

Idea

- Specify the product very properly from the beginning
- Put it in a week-by-week plan
- Deliver at the end in time and budget

Works well for construction, not for software ☺

Typical data science process



Today: Mostly agile development methods such as Scrum

Data Scientists are part of a scrum team

Aim: Quick product increments that can be demonstrated to the users

Cognitive Biases

Cognitive Biases

Cognitive biases are irrational errors of decision-making

Information Human Brain Decision

From: nobelprize.org
(Daniel Kahneman)

The conscious part of our brain can focus on one thing at a time

-> The brain creates shortcuts to make decisions. These are often erroneous

Why do we speak about cognitive biases?

As a data scientist it is your job to

- Select the right information/data to process
- Decide on how to run analyses
- Decide on how to interpret results
- Present (partly) interpreted results to stakeholders
- Decide on new directions when stuck

In all these steps, cognitive biases will affect you.

Being aware of these biases and reducing them will help to become a better data scientist

Selected Cognitive Biases

1. Confirmation Bias
2. Anchoring Bias
3. Unconscious Bias
4. Priming Bias
5. Hindsight Bias
6. Self-serving Bias
7. Status Quo Bias
8. Framing Bias
9. Automation Bias
10. Belief Bias
11. IKEA Bias
12. Information Bias

Cognitive Bias of the week: Confirmation bias

TECHNISCHE UNIVERSITÄT DARMSTADT

"Search for and interpret information in such a way that confirms pre-existing beliefs"

ORIGINAL INVESTIGATION

ONLINE FIRST

Red Meat Consumption and Mortality

Results From 2 Prospective Cohort Studies

An Pan, PhD; Qi Sun, MD, ScD; Adam M. Bernstein, MD, ScD; Matthias B. Schulze, DrPH; JoAnn E. Manson, MD, DrPH; Meir J. Stampfer, MD, DrPH; Walter C. Willett, MD, DrPH; Frank B. Hu, MD, PhD

Background: Red meat consumption has been associated with an increased risk of chronic diseases. However, its relationship with mortality remains uncertain.

Methods: We prospectively observed 37 698 men from the Health Professionals Follow-up Study (1986-2008) and 83 644 women from the Nurses' Health Study (1980-2008) who were free of cardiovascular disease (CVD) and cancer at baseline. Diet was assessed by validated food frequency questionnaires and updated every 4 years.

Results: We documented 23 926 deaths (including 5910 CVD and 9464 cancer deaths) during 2.96 million person-years of follow-up. After multivariate adjustment for major lifestyle and dietary risk factors, the pooled hazard ratio (HR) (95% CI) of total mortality for a 1-serving-per-day increase was 1.13 (1.07-1.20) for unprocessed red meat and 1.20 (1.15-1.24) for processed red meat. The corresponding HRs (95% CIs) were 1.18 (1.13-1.23) and 1.21 (1.13-1.31) for CVD mortality and 1.10 (1.06-1.14) and 1.16 (1.09-1.23) for cancer mortality. We estimated that substitutions of 1 serving per day of other foods (including fish, poultry, nuts, legumes, low-fat dairy, and whole grains) for 1 serving per day of red meat were associated with a 7% to 19% lower mortality risk. We also estimated that 9.3% of deaths in men and 7.6% in women in these cohorts could be prevented at the end of follow-up if all the individuals consumed fewer than 0.5 servings per day (approximately 42 g/d) of red meat.

Conclusions: Red meat consumption is associated with an increased risk of total, CVD, and cancer mortality. Substitution of other healthy protein sources for red meat is associated with a lower mortality risk.

Arch Intern Med.
Published online March 12, 2012.
doi:10.1001/archinternmed.2011.2287

"World Health Organization classifies red meat as "probably carcinogenic" to humans."

"Eating red meat not linked to increased risk in cancer. Enjoy your steak"

Cognitive Bias of the week: Confirmation bias

TECHNISCHE UNIVERSITÄT DARMSTADT

"Search for and interpret information in such a way that confirms pre-existing beliefs"

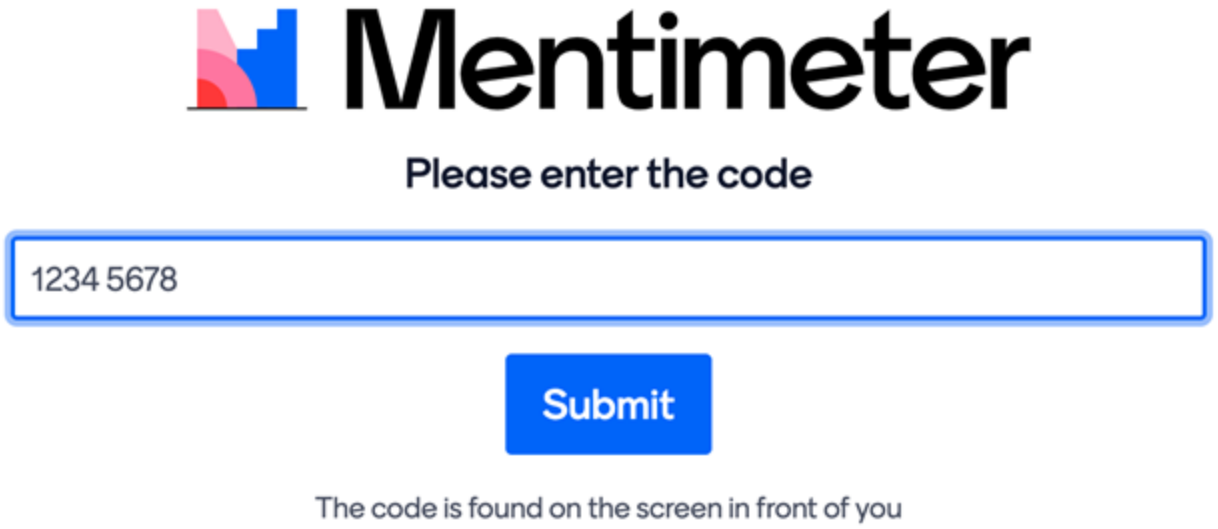
In the data science world:

- Have a higher trust in an algorithm if the result confirms a belief
- Assume bugs when the algorithm provides results that are against my belief
- Trusting sources that are aligned with my beliefs
- Make decisions based on alignment with my belief, for example
  - "We need more data and better algorithms" if I don't agree with the result
  - "This is good enough" if I agree with the result

What should I know by now?

1. Data types: Categorical, numerical, ordinal
2. How and why to do one-hot-encoding
3. How to use loc and iloc in dataframes
4. What are characteristics of bad and good plots?
5. What to keep in mind when interpreting mean values
6. How and why to do stratification
7. Groupby, merging and pivoting
8. Explain a typical data science process
9. What are cognitive biases
10. Explain the confirmation bias with an example

Mentimeter


Mentimeter
Please enter the code
1234 5678
Submit
The code is found on the screen in front of you